

Marcus Rockel

Postdoctoral Researcher, Department of Quantitative Finance, University of Freiburg

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Positions

since 01/2026	Postdoctoral Researcher , Department of Quantitative Finance, University of Freiburg
07/2021–12/2025	Doctoral Researcher , Department of Quantitative Finance, University of Freiburg
since 03/2022	Freelance Data Engineer Data migration and integration: extraction, validation, fuzzy deduplication and record linkage; LLM-based classification and entity matching on local GPU infrastructure
10/2018–06/2021	(Senior) Technical Consultant , NTT Data DACH Data analytics and big data engineering

Education

2021–2025	Dr. rer. nat. in Mathematics , University of Freiburg Thesis: <i>Chatterjee's rank correlation and classical concordance: copula perspectives, bounds, and computation</i>
2016–2018	M.Sc. in Mathematics , University of Bonn Thesis: <i>Dimensional improvements of functional inequalities for log-concave distributions</i> (advisor: Matthias Erbar)
2012–2016	B.Sc. in Mathematics , University of Bonn

Research interests

Copulas and dependence modeling; exact regions and sharp inequalities between rank correlations and dependence measures; Chatterjee's rank correlation and measures of functional dependence; positive dependence and stochastic orderings; optimal transport methods for extremal problems; formal verification of statistical results in Lean 4.

Selected articles

- The exact region and an inequality between Chatterjee's and Spearman's rank correlations
J. Ansari, M. Rockel. *Journal of Multivariate Analysis* 214, 105630, 2026.
- On the exact region between Chatterjee's rank correlation and Spearman's footrule
M. Rockel. *Journal of Computational and Applied Mathematics*, 117466, 2026.
- Dependence properties of bivariate copula families
J. Ansari, M. Rockel. *Dependence Modeling* 12(1), 20240002, 2024.
- An empirical study on new model-free multi-output variable selection methods
J. Ansari, E. Lütkebohmert, M. Rockel. SMPS 2024, *Advances in Intelligent Systems and Computing*, 9–17, Springer, 2024.

Selected preprints

- The exact Spearman rho-footrule region via optimal transport with applications to finite rankings, mixability, and Chatterjee's rank correlation
J. Ansari, M. Rockel. arXiv:2608.20176, 2026.
- Kendall and Spearman bounds for Chatterjee's rank correlation under positive dependence
M. Rockel. arXiv:2606.22074, 2026.
- Measures of association for approximating copulas
M. Rockel. arXiv:2505.08045, 2025.
- Full list of publications and preprints: marcusrockel.github.io*

Software and formal verification

lean-verifications	Lean 4 formalizations of the main results of my articles, one supplement per paper, checked in CI (corrram.github.io/lean-verifications).
copula (Lean 4)	Reusable Lean 4 library of copula mathematics underlying the supplements.
copul (Python)	Open-source package for computing with bivariate copulas and dependence measures (PyPI).

Teaching

since 2023	Python course for economists (lecture), University of Freiburg
2022–2023	Statistics (exercise sessions), University of Freiburg
2021–2023	Mathematical Methods for Economics and Finance (exercise sessions), University of Freiburg
2013–2017	Student tutor, Mathematical Institute and Institute for Applied Mathematics, University of Bonn

Skills

Programming	Python, R, Julia, Matlab, SQL, Java/Scala; Lean 4; $\text{\LaTeX}$
Languages	German (native), English (fluent)